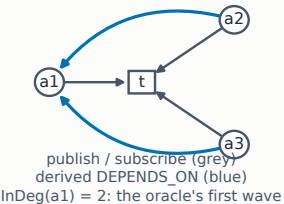
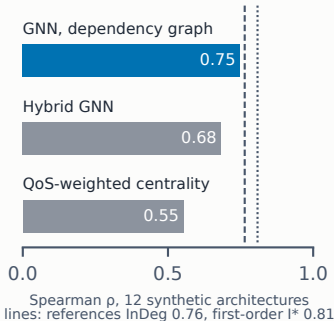


1. Derive dependencies



2. Predictors vs. references



3. Beyond one simulator

Dependency counts restate the simulator: references, not predictors.

Registered primary contrast null; hybrids beat centrality on 11/12.

Multi-criteria: centrality ranks above every learned engine (0.70 vs ≤ 0.59).

80% of the critical set needs the top 40% by the best GNN.

One simulation: seconds; GNN feature extraction: $\sim 5.6\times$ that.